

Verifier-visible interface



k slots

public-input vector
of size k

k varied
under control

Verifier operation structure (tested snarkjs templates)

Groth16

fixed:
pairing
check

per input:
EC mul + add

PLONK

fixed:
EC work +
pairing check

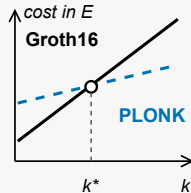
per input:
field ops,
hashing

Metering specification (regime E)



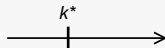
priced components:
EC precompiles,
calldata, execution,
hashing

Backend-relative cost lines in E



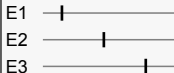
intercept $\leftarrow$ fixed block
slope $\leftarrow$ per-input block

Regime-specific crossover k^*



Groth16 cheaper **PLONK cheaper**

k^* by regime:



Explanation level

structure $\times$ metering $\rightarrow$ ranking and crossover k^*

Prediction level

structure + source-level prices $\rightarrow$ held-out native cost
tested against frozen tolerance τ